

116 Advanced Statistics for Neuroscience

Exercise 04

Pass-Fail Bernoulli GLM

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Goals of statistical analysis

Consider a class of $N = 100$ students, indexed $i = 1, 2, \dots, 100$, about which you have the following information:

- ▶ Pass/fail test results, $R_i \in \{0, 1\}$, in a vector $\mathbf{R} (N \times 1)$
- ▶ Hours of independent study, $S_i \in [0, 40]$, in a vector $\mathbf{S} (N \times 1)$
- ▶ Lectures attended, $A_i \in [0, 12]$, in a vector $\mathbf{A} (N \times 1)$

You want to establish the respective effects (if any) on test results of independent study and of lecture attendance.

To this end, you model the **log odds** of passing or failing the test as a linear function of study hours and lecture attendance. The fitted parameter values will reveal the answer to your question!

Logistic regression

With a probability π_i of passing, and $1 - \pi_i$ of failing, the **log odds** λ_i (logarithm of the ratio of probabilities) for student i are

$$\lambda_i \equiv \log \left(\frac{\pi_i}{1 - \pi_i} \right)$$

How many parameters does a linear model for two factors (hours of study, lecture attendance) require? The linear model in our case is

$$\lambda_i = \alpha_0 + \alpha_1 S_i + \alpha_2 A_i$$

Logistic regression

A programme for logistic regression typically requires a vector $\mathbf{y}$ of observations, a design matrix $\mathbf{X}$, and a parameter vector $\boldsymbol{\beta}$ with initial guesses for the parameter values. For the present problem, these items and the associated dimensions are

$$\mathbf{y} = \mathbf{R} (N \times 1), \quad \mathbf{X} = [\mathbf{n}, \mathbf{S}, \mathbf{A}] (N \times 3), \quad \boldsymbol{\beta} = \begin{pmatrix} \mu \\ \alpha_1 \\ \alpha_2 \end{pmatrix} (3 \times 1)$$

Parameter μ represents the overall pass probability, parameter α_1 the effect of study hours, and parameter α_2 the effect of lecture attendance.

Logistic regression

The binomial probability for obtaining outcome $y_i \in \{0, 1\}$, given probability π_i

$$P(y_i|\pi_i) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$$

Using the logit function $\lambda_i = \log\left(\frac{\pi_i}{1-\pi_i}\right)$ to replace $\frac{\pi_i}{1-\pi_i}$ and $(1 - \pi_i)$ with expressions of λ_i , the logarithm of this probability becomes

$$\log P(y_i|\lambda_i) = y_i \lambda_i - \log(1 + e^{\lambda_i})$$

Finally, logarithm of the joint probability of the observed outcomes over all students i is

$$\log \prod_i P(y_i|\lambda_i) = \sum_i \log P(y_i|\lambda_i) = \sum_i \left[y_i \lambda_i - \log(1 + e^{\lambda_i}) \right]$$

Perform logistic regression

Write a Python or Matlab program to determine parameters β by minimizing the **negative** likelihood of the observed outcome. First write a function to compute the negative log likelihood, from the parameter values *beta*, the design matrix *X*, and the observed outcome *y*:

```
neg_log_likelihood(beta, X, y)
```

Then find the *beta* parameters that minimize this likelihood:

```
result = minimize(neg_log_likelihood, beta_guess, args=(X, y),  
method='BFGS')
```

```
beta_fit = result.x
```

Visualization

Use the fitted model to predict the pass probability π_i of each student from his or her study hours S_i and lecture attendance A_i :

$$\pi_i = \frac{1}{1 + e^{-\lambda_i}}, \quad \lambda_i = \mu + \alpha_1 S_i + \alpha_2 A_i$$

Sort the list of π_i values from the lowest to the largest and form 10 groups of similar size, such that the lowest values are in the first group, the next lowest values in the second group, and so on.

For each group, obtain the average value of the predictor π_i and of the observables R_i (pass/fail), S_i (study hours), and A_i (lecture attendance).

Plot the observable averages as a function of the predictor averages!

Interpretation

Suppose that the logistic regression produces the result $\alpha = [-5, 0.1, 0.4]$. How do you interpret these values? How do the log-odds change with every additional hour of independent study? How do they change with every additional attended lecture?

How do the probabilities change? Starting from $\pi = 0.5$, what is the pass probability after two further hours of study or two further attended lectures?

You may want to consult the following table:

λ	+1.0	+0.9	+0.8	+0.7	+0.6	+0.5	+0.4	+0.3	+0.2	+0.1	
π	0.73	0.71	0.69	0.67	0.65	0.62	0.60	0.57	0.55	0.53	
λ	0	-0.1	-0.2	-0.3	-0.4	-0.5	-0.6	-0.7	-0.8	-0.9	-1.0
π	0.50	0.48	0.45	0.43	0.40	0.38	0.35	0.33	0.31	0.29	0.27